From Free to Fee

How Emission Permit Allocation Affects Firms

Marie Alder Eva Franzmeyer Benjamin Hattemer

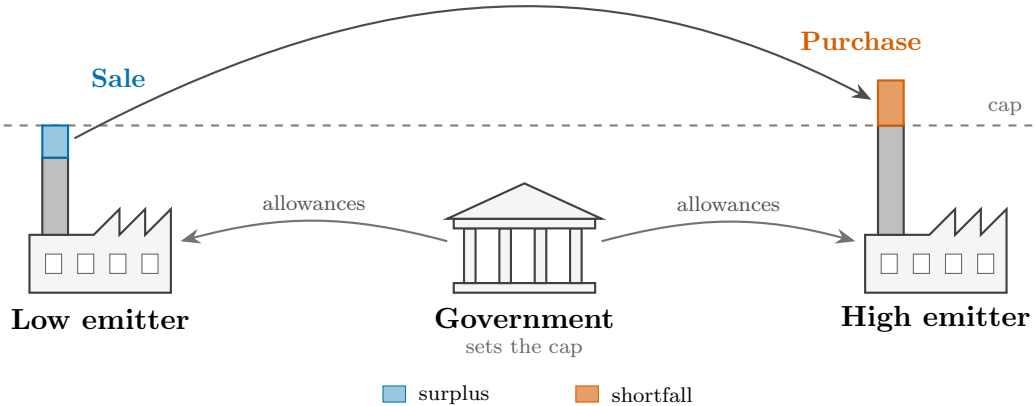
VATT Seminar

October 2026

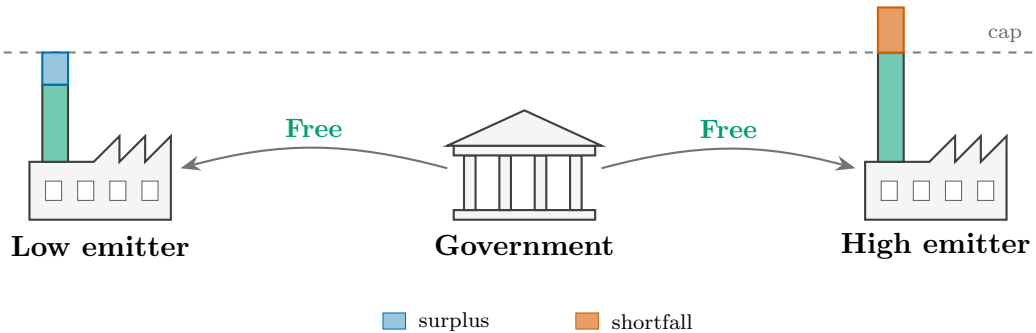
Motivation

- Global emissions covered by ETS increased to almost 20% (World Bank, 2025)
- Policy makers rely heavily on **free emission permits** (Grubb et al., 2022)
- **Benefit:** Lower costs, protection from competition (Martin et al., 2014)
- **Cost?** Depends on the ***Independence Property*** (Montgomery, 1972)
  - Under well-defined conditions, allocation does **not** affect emissions
  - If it fails: distorted abatement, higher cost (Hahn and Stavins, 2011)
- In this paper:
  1. **Empirical evidence:** Fewer permits decrease emissions and economic activity
  2. **Consistent mechanism:** Multi-product general equilibrium model

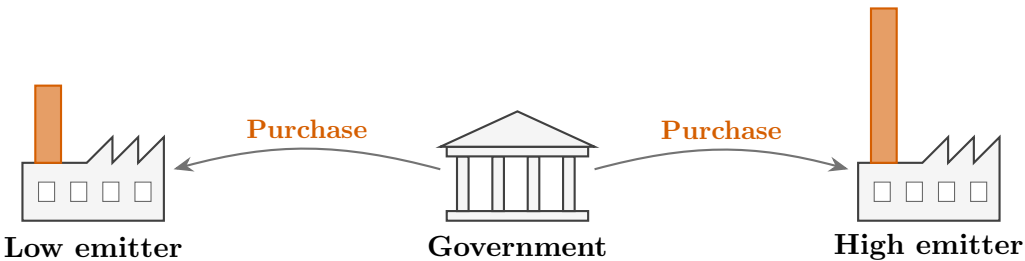
Emissions Trading System



Allocation Mechanism - Free Allocation



Allocation Mechanism - Selling to Firm



The Paper

Question

What is the effect of free emission permit allocation on firms?

Setting and Methods

- Policy: EU ETS reform moved the eligibility cutoff for free permits (announced 2018, implemented 2021)
- Manufacturing firm-level panel data (2014-2022) for DiD estimation
- Discuss mechanism with multi-product GE model based on Melitz (2003)

Results

1. **Emissions:** Substantial decrease at announcement and implementation
2. **Economic activity:** Similar decrease, but only at implementation
3. **Mechanism:** Product exit and earlier clean technology adoption

Contribution

1. **Causal effect of free permits on firm emissions**
2. **Theory: failure of the independence property**
3. **The effects of ETS on firms**

Contribution

1. **Causal effect of free permits on firm emissions**
   - Zaklan (2023): No effect for power sector in EU ETS
   - Fowlie and Perloff (2013): No effect in sub-national ETS (RECLAIM)

   ↪ This paper: Manufacturing in a large ETS with heterogeneous products

2. **Theory: failure of the independence property**

3. **The effects of ETS on firms**

Contribution

1. **Causal effect of free permits on firm emissions**

2. **Theory: failure of the independence property**
   - Transaction costs, market power, uncertainty, behavior or output-based updating (Hahn and Stavins, 2011)
   - Allocation based on operating: affects entry and exit of firms (Fischer, 2001; Ellerman, 2008; Konishi and Tarui, 2015; Fowlie et al., 2016)
   - We combine those insights with the multi-product firm literature (Eckel and Neary, 2010; Bernard et al., 2011; Mayer et al., 2014)

   ↪ This paper: Intensive margin through product exit within firms

3. **The effects of ETS on firms**

Contribution

1. **Causal effect of free permits on firm emissions**

2. **Theory: failure of the independence property**

3. **The effects of ETS on firms**
   - Literature compares regulated to unregulated firms
   - ETS reduces emissions of regulated firms
     (Bayer and Aklin, 2020; Dechezleprêtre et al., 2023; Colmer et al., 2025)
   - Emission reduction without reduced economic performance
     (Marin et al., 2018; Löschel et al., 2019; Hintermann et al., 2020; Colmer et al., 2025)

↪ This paper: (1) Regulation fixed, only free allocation varies
(2) No-tradeoff result may not persist with fewer free permits

Free Allocation Rule in the EU ETS

$$x_{ist} = \underbrace{B_s}_{\substack{\text{emission} \\ \text{benchmark}}} \cdot \underbrace{HAL_{i\bar{t}}}_{\substack{\text{historical} \\ \text{output}}} \cdot \underbrace{R_t}_{\substack{\text{cap} \\ \text{correction}}} \cdot \underbrace{CLEF_{st}}_{\substack{\text{carbon leakage} \\ \text{exposure factor}}}$$

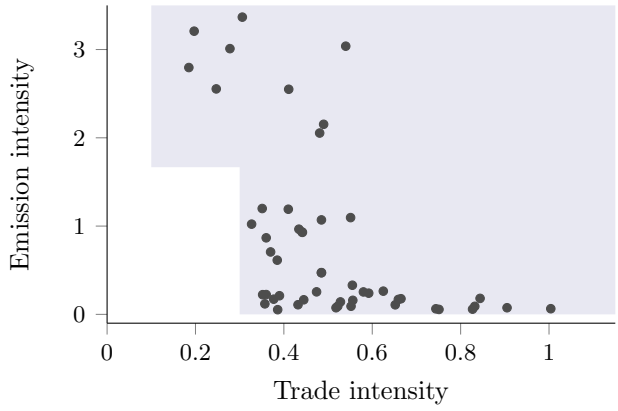
- Calculated per **sub-installation**: per **product**
- **Close a (sub)installation ⇒ lose its free permits**
- $CLEF_{st}$: 100% on carbon leakage list, 30% off it
- **Reform:** new carbon leakage list announced 2018, applied from 2021

▸ Example

Identification

Before Reform:

1. Emission intensity $\geq 5\%$ & Trade intensity $> 10\%$
2. Emission intensity $\geq 30\%$
3. Trade intensity $> 30\%$



▸ Definitions EI and TI

Identification

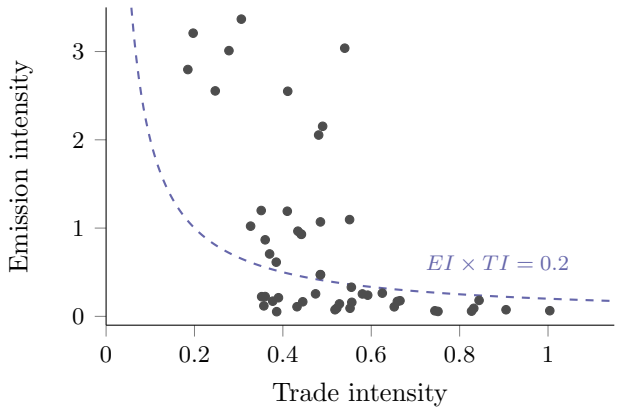
Before Reform:

1. Emission intensity $\geq 5\%$ & Trade intensity $> 10\%$
2. Emission intensity $\geq 30\%$
3. Trade intensity $> 30\%$

After Reform:

- Emission intensity $\times$ Trade intensity > 0.2

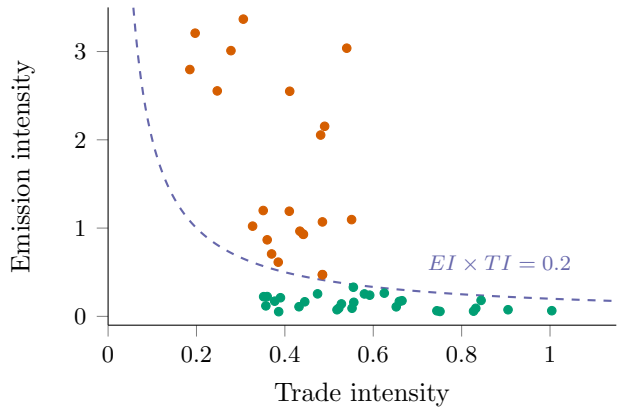
▸ Definitions EI and TI



Identification

- Control group:
  - Before: **Free**
  - After: **Free**
- Treatment group:
  - Before: **Free**
  - After: **Pay**

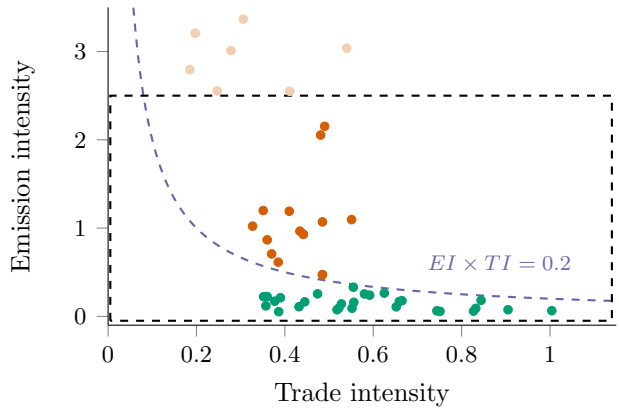
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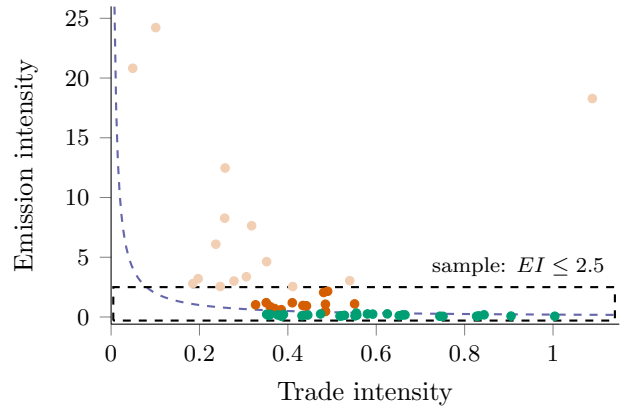
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Data

Sample: Firm-by-year balanced panel from 2014 to 2022

1. **Emissions**
2. **Economic Performance**
3. **Treatment Status**

Data

Sample: Firm-by-year balanced panel from 2014 to 2022

1. **Emissions**
   - **Installation-by-year**
   - Source: EUTL, harmonized by Jan Abrell (EUETS.INFO)
   - All stationary installations regulated under the EU ETS
   - NACE 4 sector; Orbis identifiers from JRC matching
2. **Economic Performance**
3. **Treatment Status**

Data

Sample: Firm-by-year balanced panel from 2014 to 2022

1. **Emissions**
2. **Economic Performance**
   - **Firm-by-year**
   - Source: Orbis Europe database
   - Manufacturing firms in EU member states
   - Revenue, total assets, and employees
3. **Treatment Status**

Data

Sample: Firm-by-year balanced panel from 2014 to 2022

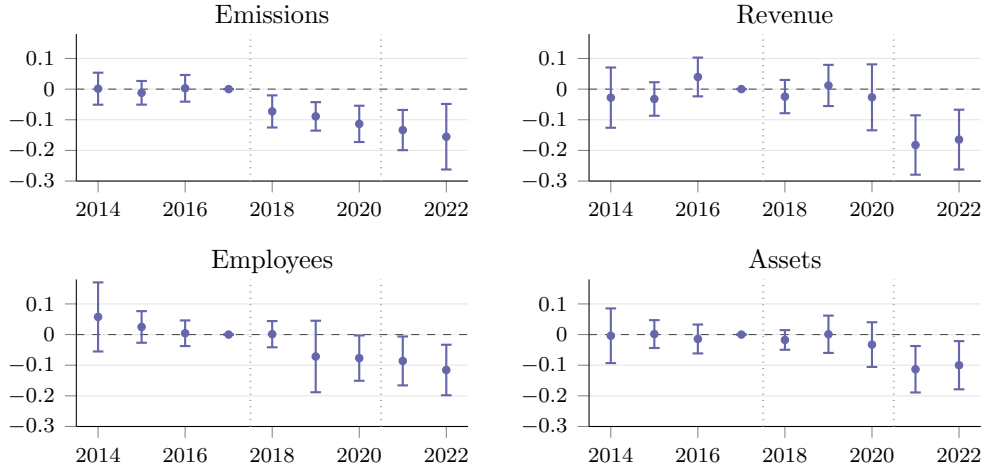
1. **Emissions**
2. **Economic Performance**
3. **Treatment Status**
   - **Sector-by-phase**
   - Source: European Commission
   - Carbon leakage lists 2015–2020 and 2021–2030, NACE 4
   - Official emission and trade intensity

Summary Statistics

	Control		Treatment	
	Mean	SD	Mean	SD
Financial variables				
Revenue (EUR thousand)	580,901	1,230,604	2,495,986	9,031,196
Employees	944	3,676	8,097	38,656
Assets (EUR thousand)	502,534	1,100,323	3,434,308	13,641,825
Emissions				
Emissions (tCO_2)	125,456	317,161	32,546	42,010
Free / Emissions, 2020 (%)	121	98	106	85
Free / Emissions, 2021 (%)	99	82	**50**	55
Sectors / firms / installations	15 / 307 / 438		30 / 117 / 177	

Balanced firm panel 2014–2022, firms from 21 countries. Firm-years: 2,246 control / 884 treated (financial), 2,763 / 1,053 (emissions)

Main Results



▸ Table ▸ Specification ▸ Trade ▸ EUA prices

Are the Groups Comparable?

		Emissions	Revenue	Employees	Assets
Baseline: EI $\leq$ 2.5 (15 sectors)	2018	-0.090^{***}	-0.012	-0.066^{*}	-0.012
	2021	-0.053^{**}	-0.161^{***}	-0.051^{**}	-0.091^{***}
Matched DiD, full sample	2018	-0.100^{***}	0.005	-0.012	0.015
	2021	-0.080^{**}	-0.117^{***}	-0.012	-0.069^{**}
Narrow window: EI $\leq$ 1.0 (8 sectors)	2018	-0.073^{**}	-0.011	-0.067	0.017
	2021	-0.056^{*}	-0.165^{***}	-0.050	-0.114^{***}
Wide window: EI $\leq$ 8.0 (25 sectors)	2018	-0.090^{***}	-0.015	-0.054^{*}	-0.035
	2021	-0.072^{***}	-0.150^{***}	-0.049^{***}	-0.075^{***}

- **Common support:** trimming to the other group's range changes nothing

$\Rightarrow$ Emissions, revenue and assets robust; employment not in the matched sample

▸ Matching ▸ Common support

Trends and Shocks?

		Emissions	Revenue	Employees	Assets
Baseline	2018	-0.090^{***}	-0.012	-0.066^{*}	-0.012
	2021	-0.053^{**}	-0.161^{***}	-0.051^{**}	-0.091^{***}
2-digit sector trends	2018	-0.123^{***}	0.013	-0.042	0.003
	2021	-0.077^{***}	-0.141^{***}	-0.033	-0.080^{***}
4-digit sector trends	2018	-0.068^{**}	-0.040	0.018	-0.002
	2021	-0.037	-0.186^{***}	0.010	-0.080^{***}
Country × year FE	2018	-0.083^{***}	-0.004	-0.050	-0.003
	2021	-0.056^{**}	-0.183^{***}	-0.049^{**}	-0.095^{***}

- **COVID and Ukraine:** no jumps in 2020/22; robust without exposed sectors

⇒ A confounder would need our timing: emissions first, activity at implementation

▸ Sector trends ▸ Country × year ▸ COVID and Ukraine

Model

- **Environment:**
  - General equilibrium model based on Melitz (2003)
  - Nested CES structure: Every firm j consists of many product lines
  - Product lines differ only in their productivity φ
  - Free allowances enter as a lump-sum transfer decreasing fixed cost

- **Key Mechanism:**
  1. **Implementation:**
     Free allowances $\downarrow$ fixed cost $\uparrow \Rightarrow$ Exit of low φ products $\Rightarrow$ firms shrink
  2. **Announcement:**
     Future revenue of surviving products $\uparrow \Rightarrow$ adopt clean technology

Demand

- CES utility function over a continuum of goods:

$$U = \left(\sum_{j=1}^{N} Q_j^{\frac{\varepsilon-1}{\varepsilon}} \right)^{\frac{\varepsilon}{\varepsilon-1}} \quad \text{where} \quad Q_j = \left(\int_{\Omega_j} q_j(\varphi)^{\frac{\sigma-1}{\sigma}} \, d\varphi \right)^{\frac{\sigma}{\sigma-1}}$$

- Ω_j is the mass of goods of firm j, indexed by productivity φ
- Set of varieties can be understood as an aggregated good $Q = U$ with price P
- Elasticity of substitution larger within than across firms: $\sigma > \varepsilon > 1$

Production

- Production of a **product line** requires labor and energy
- Each product line draws a productivity level φ_j from $g(\varphi)$
- Lump-sum free allowances (x_j) lower the fixed cost per period: $F_j = f_{\text{fix}} - x_j\tau$

$$\min_{\ell,e} \; w\ell + (P_e + \tau)e \quad \text{s.t.} \quad \varphi\ell^{\alpha}e^{1-\alpha} = 1$$

- Minimizing cost yields the **unit cost** $c_u(\varphi)$ and **pricing rule** $p_j(\varphi)$:

$$c_u(\varphi) = \frac{1}{\varphi}\underbrace{\left(\frac{w}{\alpha}\right)^{\alpha}\left(\frac{P_e+\tau}{1-\alpha}\right)^{1-\alpha}}_{\equiv\Psi}, \qquad p_j(\varphi) = \underbrace{\frac{\sigma}{\sigma-1}}_{\text{constant markup}}\frac{\Psi}{\varphi}$$

$\Rightarrow$ **Pricing independent of** F_j

Aggregation

- The aggregate price of firm j is: $P_j = M_j^{\frac{1}{1-\sigma}} p_j(\tilde{\varphi}_j)$
  - M_j: total mass of product lines of firm j
  - $p_j(\tilde{\varphi}_j)$: pricing rule using CES weighted average productivity $\tilde{\varphi}_j$
- Relative to the economy-wide aggregate we get:

$$\left(\frac{P_j}{P}\right)^{1-\varepsilon} = \frac{R_j}{R} = \frac{L_j^p}{L^p} = \frac{E_j^p}{E^p}, \qquad \text{where} \qquad \frac{P_j}{P} = \frac{M_j^{-\frac{1}{\sigma-1}} \tilde{\varphi}_j^{-1}}{\left(\sum_m M_m^{\frac{\varepsilon-1}{\sigma-1}} \tilde{\varphi}_m^{\varepsilon-1}\right)^{-\frac{1}{\varepsilon-1}}}$$

Policy: Implementation

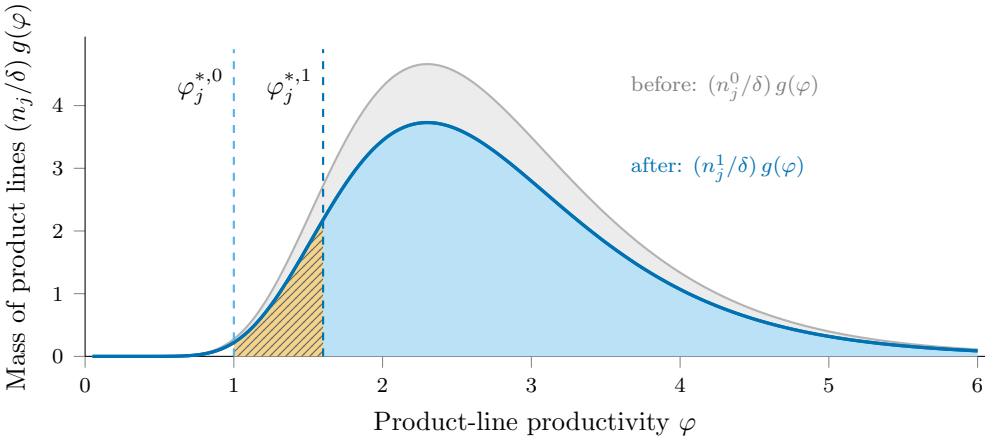
- Policy experiment: $x_j \downarrow \Rightarrow F_j = f_{\text{fix}} - x_j\tau \uparrow$:

$$\frac{\partial}{\partial F_j} \ln\left(\frac{P_j}{P_{-j}}\right) = \Bigg[\underbrace{-\frac{1}{\sigma-1}\frac{1}{M_j}\frac{\partial M_j}{\partial F_j}}_{\text{selection and entry } (+)} \quad \underbrace{-\frac{1}{\tilde{\varphi}_j}\frac{\partial \tilde{\varphi}_j}{\partial F_j}}_{\text{composition } (-)} \Bigg] \quad (+\,\text{GE spillovers})$$

- Pareto productivity with $\theta > \sigma - 1$: selection and entry dominate composition
- **Implementation:** $x_j \downarrow \Rightarrow F_j \uparrow \Rightarrow M_j \downarrow \Rightarrow \frac{P_j}{P_{-j}} \uparrow \Rightarrow \frac{R_j}{R_{-j}} \downarrow \Rightarrow \frac{L_j^p}{L_{-j}^p} \downarrow \Rightarrow \frac{E_j^p}{E_{-j}^p} \downarrow$

▸ Details Channel

Policy: Implementation



Technology

- Cleaner technology at fixed cost K: lower emission factor of energy $\kappa_h < \kappa_\ell$

$$\min_{\ell,e} \; w\ell + (P_e + \tau\kappa)e \quad \text{s.t.} \quad \varphi \ell^\alpha e^{1-\alpha} = 1 \quad \Rightarrow \quad p(\kappa;\varphi) = \frac{\sigma}{\sigma-1}\frac{\Psi(\kappa)}{\varphi}$$

- Adoption increases profits by:

$$\Delta\pi(\varphi) = \frac{1}{\sigma}(m^{1-\sigma}-1) r_\ell(\varphi), \quad \text{where} \quad m \equiv \frac{\Psi(\kappa_h)}{\Psi(\kappa_\ell)} = \left(\frac{P_e + \tau\kappa_h}{P_e + \tau\kappa_\ell}\right)^{1-\alpha} < 1$$

- A product line adopts iff $\frac{\Delta\pi(\varphi)}{\delta} \geq K$:

$$\varphi_j^A = \varphi_j^* \left[\frac{\sigma\delta K}{r_\ell(\varphi_j^*)(m^{1-\sigma}-1)}\right]^{1/(\sigma-1)}$$

- Adoption lowers emissions per unit of revenue: $\frac{\kappa\, e(\varphi)}{r(\varphi)} \propto \frac{\kappa}{P_e+\tau\kappa}$

Policy: Anticipation

- **Remaining** product lines increase lifetime profits:

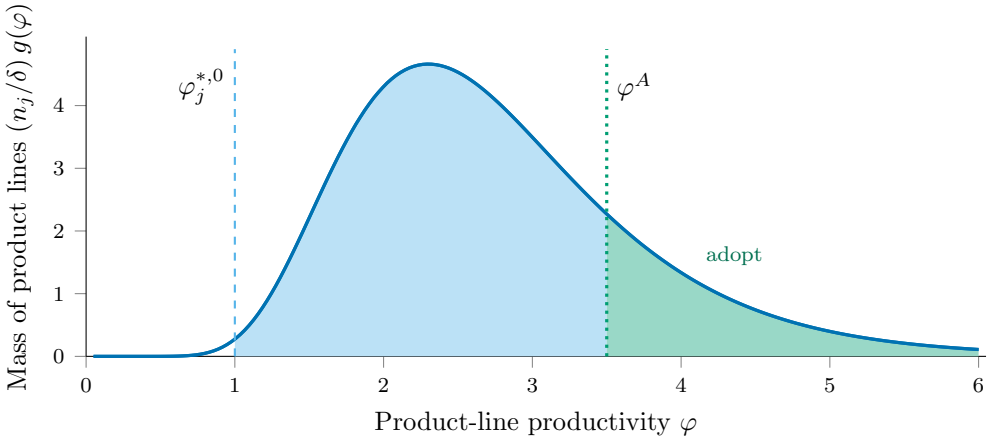
$$\begin{aligned}\Delta v^0(\varphi) &= \Delta\pi^0(\varphi) + (1-\delta)\Delta\pi^1(\varphi) + (1-\delta)^2\Delta\pi^1(\varphi) + \dots \\ &= \Delta\pi^0(\varphi) + \frac{1-\delta}{\delta}\Delta\pi^1(\varphi)\end{aligned}$$

- Anticipation effect $\Phi_j > 1$ increases technology adoption:

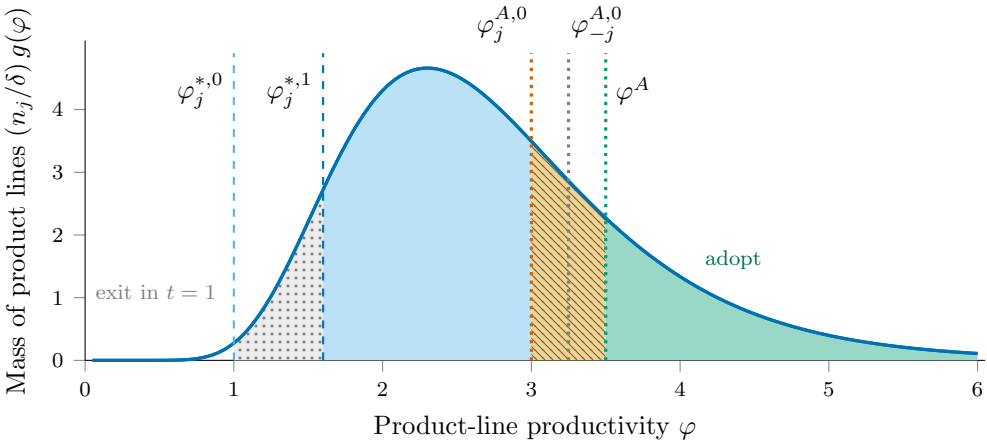
$$\varphi_j^{A,0} = \varphi_j^{*,0}\left[\frac{\sigma K}{r_\ell^0(\varphi_j^{*,0})(m^{1-\sigma}-1)} \cdot \frac{1}{1+\frac{1-\delta}{\delta}\Phi_j}\right]^{\frac{1}{\sigma-1}}, \quad \Phi_j \equiv \frac{r_\ell^1(\varphi)}{r_\ell^0(\varphi)}$$

$$\frac{\partial}{\partial F_j^1}\ln\left(\frac{\varphi_j^{A,0}}{\varphi_{-j}^{A,0}}\right)\Bigg|_{\Phi=1} = -\frac{1-\delta}{\sigma-1}(\sigma-\varepsilon)\left(\underbrace{\frac{1}{P_j^1}\frac{\partial P_j^1}{\partial F_j^1}}_{+} \underbrace{-\frac{1}{P_{-j}^1}\frac{\partial P_{-j}^1}{\partial F_j^1}}_{+}\right)$$

Policy: Anticipation



Policy: Anticipation



Conclusion

1. **Empirics:** fewer free permits
   - Emissions fall at the announcement and at the implementation
   - Economic activity falls only at the implementation
2. **Mechanism:** conditional allocation
   - Free permits are tied to operating (sub-)installations, our product lines
   - Multi-product model: this conditional allocation explains the observed pattern
3. **Implication**

$\Rightarrow$ Free permits limit leakage, but **distort the market**: weaker lines survive

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Appendix

Example Allocation Rule

Iron foundry (NACE C24.51, in our control group), allocation for 2021:

	On the list	If off the list
B_s: iron casting benchmark	0.282 allowances per t	
$HAL_{i\bar{t}}$: median output 2014–2018	100,000 t	
R_t: cap correction 2021–2025	1	
$CLEF_{st}$	1.0	0.3
x_{ist}	**28,200**	**8,460**

- Losing the status: −19,740 allowances, ≈ €1.6m per year at €80
- Treated sectors have no product benchmark: permits per heat/fuel use

Back

Carbon Leakage Indicators

$$\text{Emission intensity} = \frac{\text{direct emissions} + \text{electricity use} \times \text{emission factor}}{\text{gross value added}} \quad \left[\frac{\text{kg } CO_2}{€}\right]$$

$$\text{Trade intensity} = \frac{\text{imports} + \text{exports}}{\text{imports} + \text{production}} \quad \text{(trade with non-EU countries)}$$

- **Phase IV:** on the list if EI $\times$ TI > 0.2; qualitative assessment for 0.15–0.2
- **Phase III: carbon cost** instead of EI (emissions at €30/t over GVA): 5%, 30%
- Both published by the European Commission for NACE 4 sectors

Back

Main Results: Difference-in-Differences

$$\ln(y_{i,s,t}) = \rho_A \left(\text{Treat}_s \times \text{Post2018}_t\right) + \rho_I \left(\text{Treat}_s \times \text{Post2021}_t\right) + \alpha_i + \gamma_t + \varepsilon_{i,s,t}$$

	Emissions	Revenue	Employees	Assets
Announcement 2018 (ρ_A)	-0.090^{***}	-0.012	-0.066^{*}	-0.012
	(0.026)	(0.038)	(0.038)	(0.029)
Implementation 2021 (ρ_I)	-0.053^{**}	-0.161^{***}	-0.051^{**}	-0.091^{***}
	(0.025)	(0.032)	(0.022)	(0.022)
Observations	3,816	3,130	3,130	3,130

Back

Event Study: Specification

$$\ln(y_{i,s,t}) = \sum_{\substack{k=-4 \\ k\neq -1}}^{4} \beta_k \left(\text{Treat}_s \times \mathbb{1}[t - 2018 = k]\right) + \alpha_i + \gamma_t + \varepsilon_{i,s,t}$$

- $y_{i,s,t}$: emissions, revenue, employees or assets of firm i in sector s in year t
- $\text{Treat}_s = 1$ if sector s lost its free allocation in the reform
- k: years relative to the announcement (2018); baseline $k = -1$ (2017)
- α_i and γ_t: firm and year fixed effects
- Standard errors clustered at the sector level

Back

Trade: Specification

- **Question:** shrinking firms lower trade; carbon leakage would raise imports
- **Data:** ComExt trade, TRAINS tariffs; same sectors as firm level

$$\ln y_{ij,s,t} = \rho_A \left(\text{Treat}_s \times \text{Post2018}_t\right) + \rho_I \left(\text{Treat}_s \times \text{Post2021}_t\right) + \beta\, T_{ij,s,t} + \tau_{ij,t} + \tau_{ij,s} + u_{ij,s,t}$$

- $y_{ij,s,t}$: exports or imports of sector s between countries i and j
- $T_{ij,s,t}$: bilateral tariffs; $\tau_{ij,t}$, $\tau_{ij,s}$: pair-year and pair-sector FE

▸ Results ▸ Back

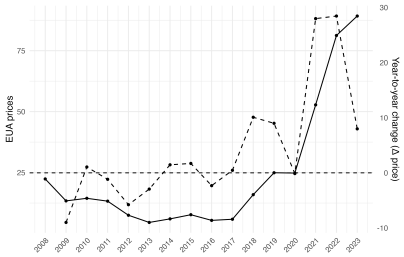
Trade: Results

Dependent Variables:	Exports (logs)	Imports (logs)
Model:	(1)	(2)
Variables		
Treatment × Post (2018)	-0.0176	0.0247
	(0.0254)	(0.0284)
Treatment × Post (2021)	-0.0461	-0.0529
	(0.0330)	(0.0495)
p-value (2018)	0.492	0.390
p-value (2021)	0.170	0.291
Fixed-effects		
Sector-i-j	Yes	Yes
Time-i-j	Yes	Yes
Fit statistics		
Observations	879,909	571,286
R^2	0.89008	0.90659
Within R^2	7.41×10^{-5}	4.44×10^{-5}

- Implementation: trade about -5%
- Symmetric: no **carbon leakage**
- **Low power:** MDEs of 9.5 and 14 log pts
- Below -16%: ETS firms only part of trade

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EUA Prices and the Reform



Solid: EUA price (EUR/t); dashed: year-to-year change. Source: EEX

- Prices: about €6 (2017) to over €80 (2022)
- **Value:** estimate includes the permits' value
- **Direct effect:** threat only if groups differ

$$\Delta CO_2 = \underbrace{\delta}_{\text{abatement}} \cdot \underbrace{E^0}_{\text{exposure}} \cdot \Delta P$$

  - **Exposure:** higher in control → conservative
  - **Abatement:** no hard-to-abate sectors

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Matching

- Methodology commonly used (Colmer et al., 2025; Zaklan, 2023)
- One-to-one nearest-neighbor matching using a Mahalanobis distance
- Matched in 2017 on emission intensity, assets, emissions and non-EU imports
- Results remain robust using:
  - No replacement
  - Euclidean distance
  - Up to 4 nearest neighbors

Matching Results

	Verified Emissions	Revenue	Employees	Assets
Treatment 2018	-0.100***	0.005	-0.012	0.015
	(0.027)	(0.027)	(0.030)	(0.024)
Treatment 2021	-0.080**	-0.117***	-0.012	-0.069**
	(0.037)	(0.035)	(0.031)	(0.029)
Observations	2875	2732	2732	2732
Adjusted R^2	0.021	0.016	-0.000	0.007

Alternative Matching Specifications: Verified Emissions

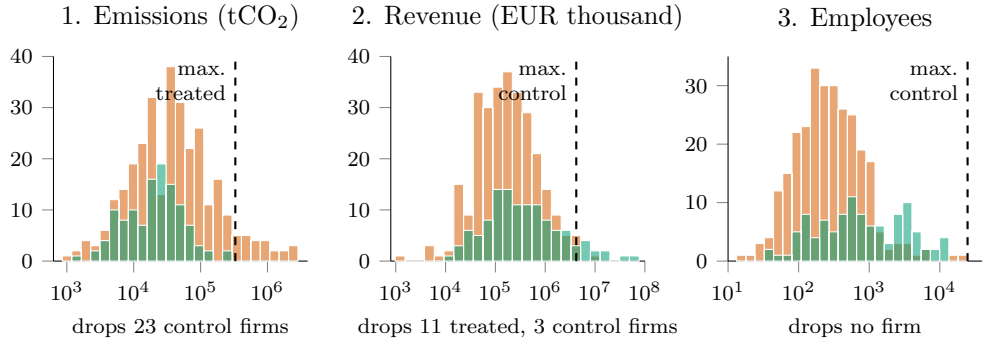
	Euclidean	No replacement	4NN	3NN	2NN
Treatment 2018	-0.111***	-0.085***	-0.113***	-0.091***	-0.091***
	(0.029)	(0.026)	(0.028)	(0.026)	(0.028)
Treatment 2021	-0.095**	-0.066*	-0.084**	-0.092**	-0.087**
	(0.039)	(0.033)	(0.037)	(0.038)	(0.043)
Observations	2875	2875	2300	1725	1150
Adjusted R^2	0.029	0.015	0.026	0.026	0.023

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Common Support: Trimming

Firm averages, cumulative trimming at the reference group's maximum: 424 → 387 firms

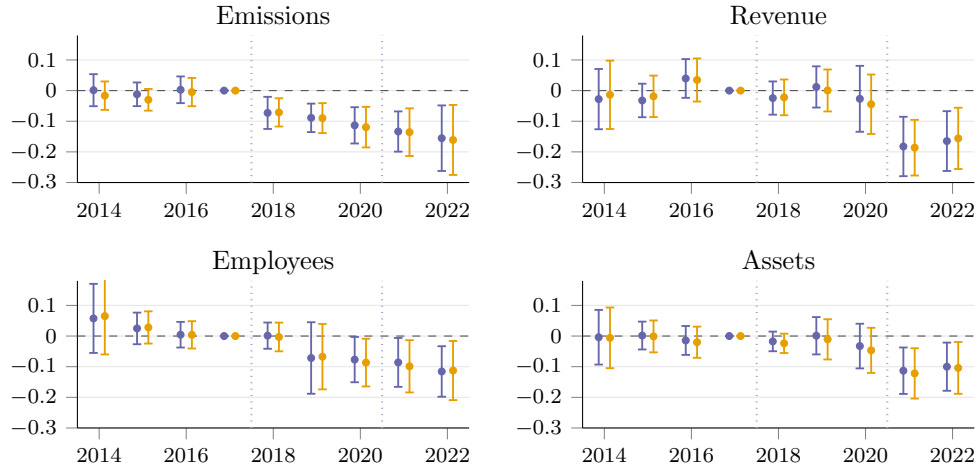
Control Treated --- maximum of the reference group (pre-treatment)



Results Back

Common Support: Results

● Baseline (424 firms) ● Trimmed sample (387 firms)



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Sector Trends

Sector trends:	No trends	2-digit trends	4-digit trends
Model:	(1)	(2)	(3)
Verified Emissions			
Treatment × Post (2018)	-0.090***	-0.123***	-0.068**
	(0.026)	(0.027)	(0.029)
Treatment × Post (2021)	-0.053**	-0.077***	-0.037
	(0.025)	(0.027)	(0.027)
Observations	3,816	3,816	3,816
Revenue			
Treatment × Post (2018)	-0.012	0.013	-0.040
	(0.038)	(0.047)	(0.035)
Treatment × Post (2021)	-0.161***	-0.141***	-0.186***
	(0.032)	(0.039)	(0.045)
Observations	3,130	3,130	3,130
Employment			
Treatment × Post (2018)	-0.065*	-0.042	0.018
	(0.038)	(0.048)	(0.040)
Treatment × Post (2021)	-0.051**	-0.033	0.010
	(0.022)	(0.025)	(0.037)
Observations	3,130	3,130	3,130
Total Assets			
Treatment × Post (2018)	-0.012	0.003	-0.002
	(0.029)	(0.027)	(0.020)
Treatment × Post (2021)	-0.091***	-0.080***	-0.080***
	(0.022)	(0.026)	(0.023)
Observations	3,130	3,130	3,130

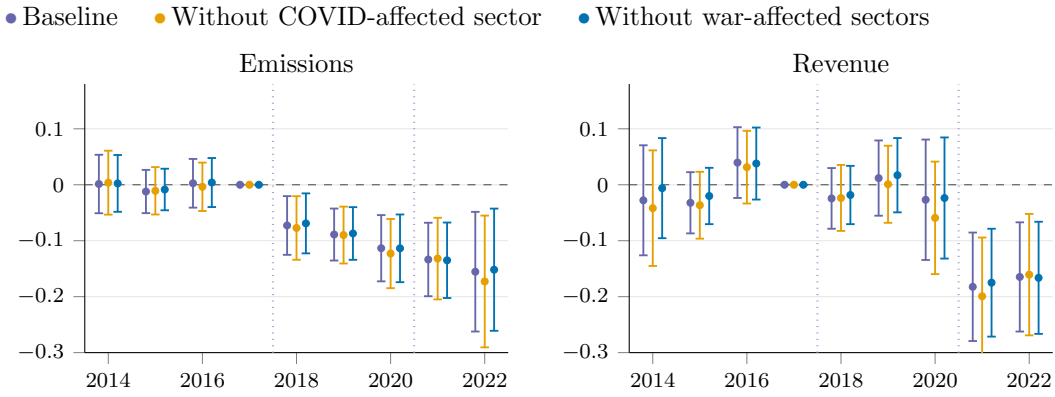
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Country × Year Fixed Effects

Dependent Variables:	Verified Emissions	Revenue	Employees	Assets
Model:	(1)	(2)	(3)	(4)
Variables				
Treatment x Post (2018)	-0.0828***	-0.0040	-0.0498	-0.0031
	(0.0278)	(0.0430)	(0.0358)	(0.0295)
Treatment x Post (2021)	-0.0560**	-0.1825***	-0.0486**	-0.0946***
	(0.0240)	(0.0313)	(0.0211)	(0.0227)
Fixed-effects				
Firm FE	Yes	Yes	Yes	Yes
Country x Time FE	Yes	Yes	Yes	Yes
Fit statistics				
Observations	3,789	3,094	3,094	3,094
R^2	0.97770	0.97674	0.97666	0.98400
Within R^2	0.01330	0.01681	0.00441	0.00675

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COVID and the Ukraine War



- **COVID-affected:** pharmaceutical preparations (C21.20)
- **War-affected:** weapons and ammunition (C25.40), shipbuilding (C30.11)

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Policy: Implementation

1. **Entry-flow channel:** Higher F_j discourages entry: $\frac{\partial n_j}{\partial F_j} < 0$
2. **Selection channel:** Higher F_j raises the survival cutoff: $\frac{\partial \varphi_j^*}{\partial F_j} > 0$
3. **Composition channel:** Remaining product lines more productive $\frac{\partial \bar{\varphi}_j}{\partial F_j} > 0$

Implementation channel M_j

- M_j is given by stationarity:

$$\underbrace{\delta M_j}_{\text{Exit}} = \underbrace{n_j \left[1 - G(\varphi_j^*)\right]}_{\text{Entry}} \Rightarrow M_j = \frac{n_j}{\delta}\left[1 - G(\varphi_j^*)\right].$$

- Then the effect of F_j on M_j is:

$$\frac{dM_j}{dF_j} = \underbrace{\frac{dn_j}{dF_j}\frac{1 - G(\varphi_j^*)}{\delta}}_{\text{Entry-flow (-)}} \underbrace{- \frac{n_j}{\delta} g(\varphi_j^*)\frac{d\varphi_j^*}{dF_j}}_{\text{Selection (-)}} < 0$$

- n_j is the inflow of product lines and φ_j^* is the cutoff productivity

- We can show that $\frac{dn_j}{dF_j} < 0$ and $\frac{d\varphi_j^*}{dF_j} > 0$ using a Pareto tail distribution

Implementation channel $\tilde{\varphi}_j$

- $\tilde{\varphi}_j$ is given by:

$$\tilde{\varphi}_j(\varphi_j^*) = \left[\frac{1}{1 - G(\varphi_j^*)} \int_{\varphi_j^*}^{\infty} \varphi_j^{\sigma-1} g(\varphi_j)\, d\varphi_j \right]^{\frac{1}{\sigma-1}}$$

- Then the effect of F_j on $\tilde{\varphi}_j$ is:

$$\frac{\partial \tilde{\varphi}_j}{\partial F_j} = \underbrace{\frac{d\tilde{\varphi}_j}{d\varphi_j^*}}_{\text{truncation raises mean (+)}} \underbrace{\frac{d\varphi_j^*}{dF_j}}_{\text{tougher cutoff (+)}} > 0$$

- We can show that $\frac{d\tilde{\varphi}_j}{d\varphi_j^*} > 0$ and $\frac{d\varphi_j^*}{dF_j} > 0$ using a Pareto tail distribution

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